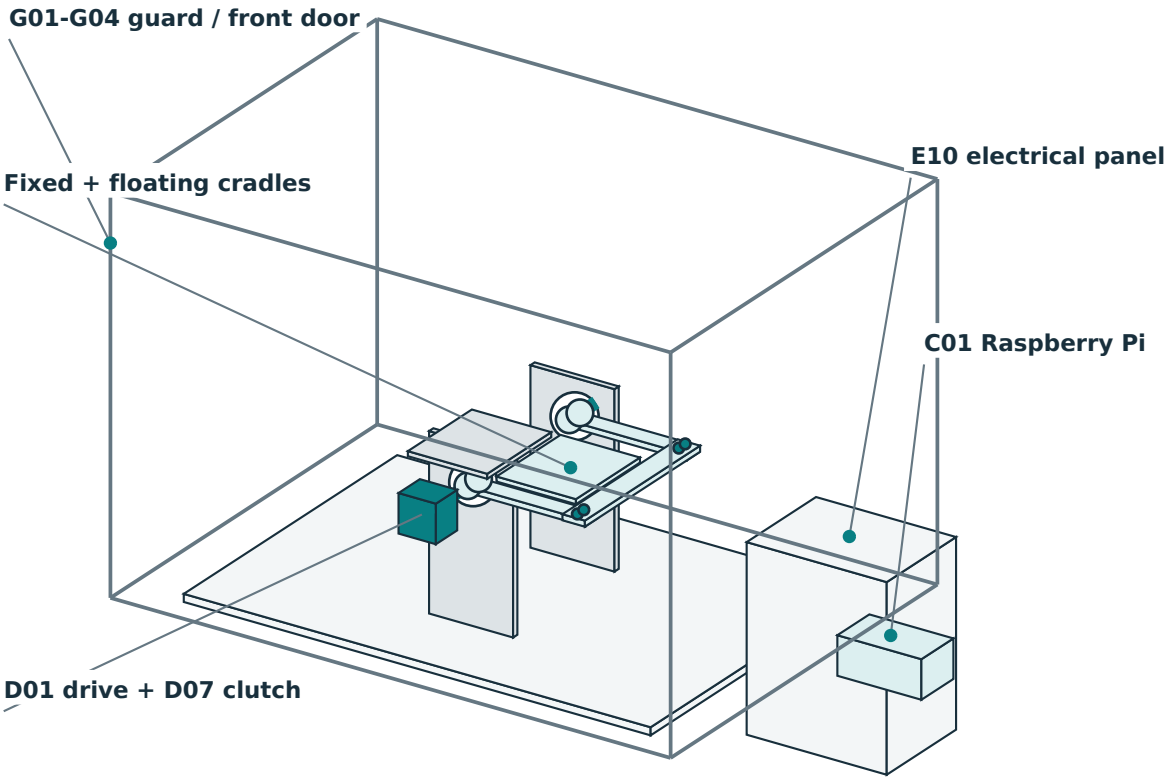
Build the bench

Illustrated engineering work instructions | 26 numbered operations



One station. From sorted parts to a supervised phone trial.

Separate from the FoldBench design-and-assembly document. This book adds assembly order, exploded views, part callouts, wire schedules and inspection records. The Python package remains foldbench.

PROTOTYPE / ENGINEERING RELEASE REQUIRED

Physical hardware has not been built or validated. Close H1-H5 before the affected work. Illustrations show assembly intent; they are not scale templates or collision-checked CAD.

Units: mm unless stated. Published 4 November 2026. Build serial: _____

READ FIRST

How to use this book

Read the entire operation before picking up a tool. The parts strip identifies the kit. Teal highlights the assembly of interest, gray is context, arrows show fit/direction, and dashed lines are alignments or signal relationships. Colour is supplementary: every critical item is labeled.

Steps	Stage	Work	Pages
01-02	Prepare	Inventory, actual specimen measurements, fabrication	5-6
03-11	Mechanics	Base, towers, yoke, drive, float, shoes, stops, damper	7-15
12-13	Guard	Frame, panels, door and external controls	16-17
14-19	Electrical	Enclosure, power, interlocks, signals and Pi	18-23
20-26	Commission	Static checks, indexing, dummy and phone trials	24-30
A-D	Bench references	Wire list, fasteners, software, signed records	31-35

Before starting

Work on a rigid, level bench. Provide access to all supply plugs and the E-stop. Keep the phone away until step 25.

Use the complete 60-line procurement BOM in hardware/BOM.md. This companion repeats the parts needed at each operation; it does not certify that provisional mounting allowances are complete.

Use caliper/angle gauge, straightedge/square, low-range torque tool, hex keys, crimp tools and meter. Add an indicator/height gauge, straight 6 mm setup rod and timing/video reference if not already available; these are additional metrology aids.

Isolate all supplies before adjusting mechanical parts or wiring. The Pi supply is independent of the motor disconnect. Restrain gravity-loaded parts before releasing a clamp/coupling.

Release before cutting

This booklet exposes gaps in the previous envelope design. An engineer must close the items below, not infer hidden dimensions.

Hold	Affected parts	Required engineering release
H1 / yoke	FAB-M03/M04, hubs, horn, bearings	Resolve arm/hub center alignment and joint edge distances. The old 190 mm crossbar spans support centers +/-95, but hole rows at +/-6 from those centers can fall beyond its ends. The arm envelope also puts outer holes too close to an edge. Release corrected blanks, joints and vendor mounting patterns before machining.
H2 / float	Rails, carriers, shoes and shoulder-screw stack	Release carrier connections, rail/carriage screws, travel retainers and bushing/spacer dimensions. Demonstrate +/-6 mm radial and approximately +/-2 mm tangential travel without losing retention.
H3 / guard	Frame joints, outriggers, panels and controls	Release load-bearing supports from 500 x 400 base to wider guard, frame overlaps, panel/door hole patterns and control mounting panel. G01 has only 480 mm stock remaining before saw kerfs after its listed frame cuts; recheck extra material and fasteners.
H4 / drive	Configured clutch and damper	Confirm EAS12 configuration and low-torque adjustment with vendor; select the actual damper and design its stationary reaction bracket and output coupling. These are not fully specified purchased kits.
H5 / electrical	Safety circuit and permit board	Approve relay revision, series-device fault behavior, fuses/inrush, device contact IDs and optocoupler package pin map. Verify actual motor-power removal and no automatic restart.

Retain signed drawings with this booklet. Blank fields are intentional measurements, not permission to guess. The assembly figures are conceptual and may show revised joint proportions; they do not supersede the released fabrication drawing. Update the stock/fastener order after H1-H4 are resolved.

Fabricated-part register

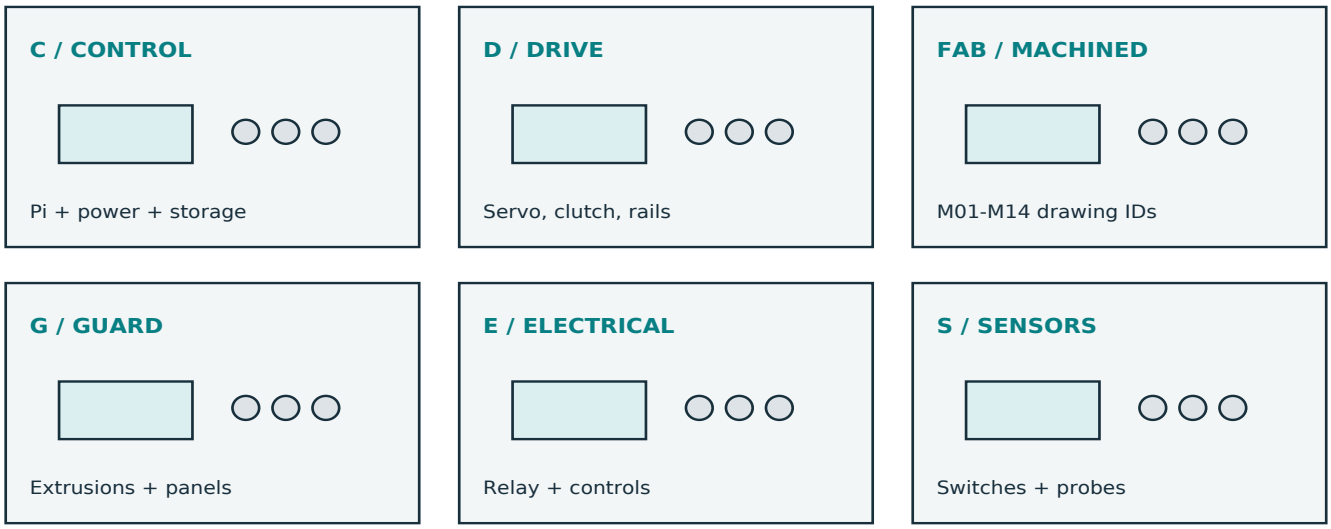
Prefix every drawing identifier with FAB. BOM-M03 is a shoe blank; FAB-M03 is a yoke arm. Never interchange them.

FAB ID	Qty	Reference stock / operation	Procurement
M01	1	Base 500 x 400 x 8; grid/foot holes: step 02	BOM M01
M02	2	Tower 100 x 180 x 8 + 100 x 50 x 6 foot; vendor adapters	BOM M02
M03	2	Arm 125 x 25 x 6 envelope; H1 revised joint required	BOM M02
M04	1	Crossbar 190 x 25 x 6 envelope; H1 revision required	BOM M02
M05	1	Moving plate 100 x 135 x 4; four 4.5 x 20 slots	BOM M02
M06	1	Fixed plate 100 x 135 x 4; measured camera relief	BOM M02
M07	2	Riser 100 x 125 x 6 + 100 x 40 x 6 foot; shim height	BOM M02
M08	8	Acetal shoe 20 x 15 x 8; 4.5 x 10 slot; measured contour	BOM M03
M09	1	Shelf 100 x 100 x 6 + two 100 mm slotted angles	BOM M02
M10	4	Stop/switch angle 40 x 25 x 3; 4.5 x 15 base slots	BOM M02
M11	1	Adapter diameter 30 x 10; 6 H7 split bore; vendor horn pattern	BOM M02
M12	2	Guide carrier 100 x 25 x 4; rail pattern and joints H2	BOM M02
M13	2	Acetal cam 20 x 15 x 3; output-side adjustable fixing	BOM M02*
M14	5	4 mm panels: 560 x 360 (2), 460 x 360 (2), 560 x 460 (1)	BOM G03

*FAB-M13 must be acetal although grouped in the plate-stock allowance. General metal: 6061-T6. Contact shoes: acetal plus 1 mm silicone. Stock sizes are Rev A references; H1/H2 may require revised blanks and additional retainers/brackets. Vendor patterns and fastener torques follow delivered parts.

Inventory and measure the specimen

PARTS BOM T01 caliper/angle gauge; T05 two-leaf dummy; all 60 BOM lines; actual phone for measurements only.
TOOLS Caliper, straightedge, camera, parts trays



Also sort W wire, F fasteners, T tools and optional O parts.

1. Sort purchased parts by BOM ID. Mark fabricated pieces FAB-M01 through FAB-M14; these are drawing IDs, not the similarly numbered BOM stock items. Keep vendor screws with their component.
2. Record each phone half's width, thickness, rigid contact lands, camera projection, USB and button keep-outs. Measure actual relaxed closed/open angles. Nominal exterior dimensions are not drilling dimensions.
3. Make a two-leaf clearance dummy from those measurements. Mark stationary USB half, moving half and hinge center. A simple dummy does not reproduce the real hinge path or torque.
4. Obtain the actual bearing, horn, hub, slide and clutch drawings. Resolve the release register on page 3 before machining their interfaces.

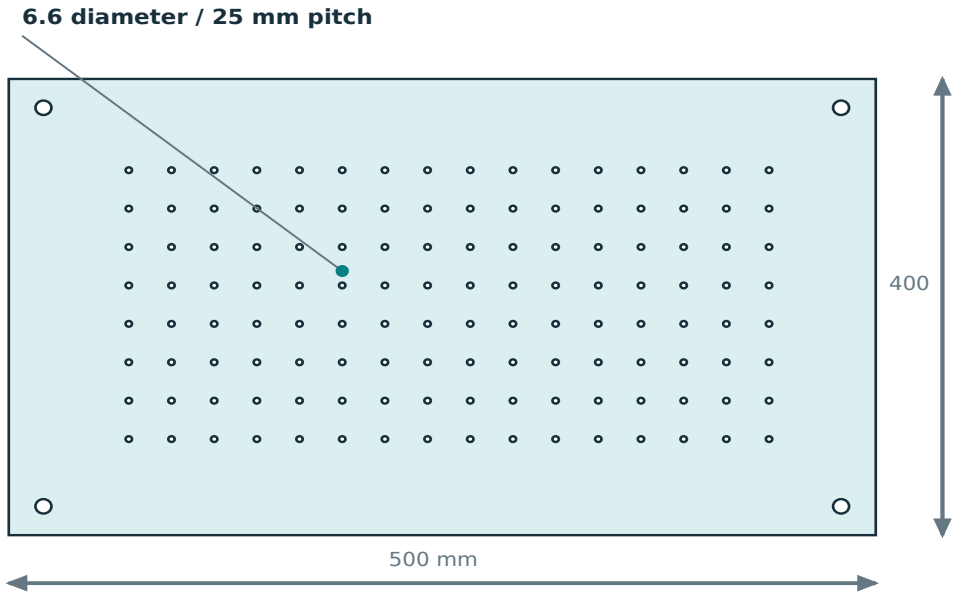
CHECK BEFORE CONTINUING

All parts identified; specimen measurement sheet completed; unresolved interfaces assigned to an engineer. No phone is installed during mechanical assembly.

Phone/dummy drawing rev: _____ Vendor drawing pack rev: _____

Prepare the fabricated parts

PARTS BOM M01 base, M02 plate-stock set, M03 shoe blanks x8; FAB-M01 to M13; F04 shims/witness paint.
TOOLS Mill/drill, deburr tools, caliper, square



FAB-M01 top view. Grid extent: 350 x 250. Schematic only.

1. Machine the base 500 x 400 x 8 mm. With origin at its center, the reference grid is X=-175 to +175 and Y=-125 to +125 in 25 mm increments, holes diameter 6.6 mm. Foot centers are X=+/-230, Y=+/-180.
2. Make plate blanks to the fabrication register. Transfer mounting patterns from accepted vendor drawings; keep adjustment slots within adequate edge distances. Do not use this booklet or the envelope SVG as a drill template.
3. Hold FAB-M03/M04 yoke drilling until the joint is corrected: the old crossbar and arm envelope dimensions do not leave reliable edge distance at all fasteners. Use a released detail drawing with two M4 fasteners per outer joint.
4. Deburr; break general edges at least 0.5 mm and round phone-adjacent corners to at least R2. Clean all chips and mark each part away from contact surfaces.

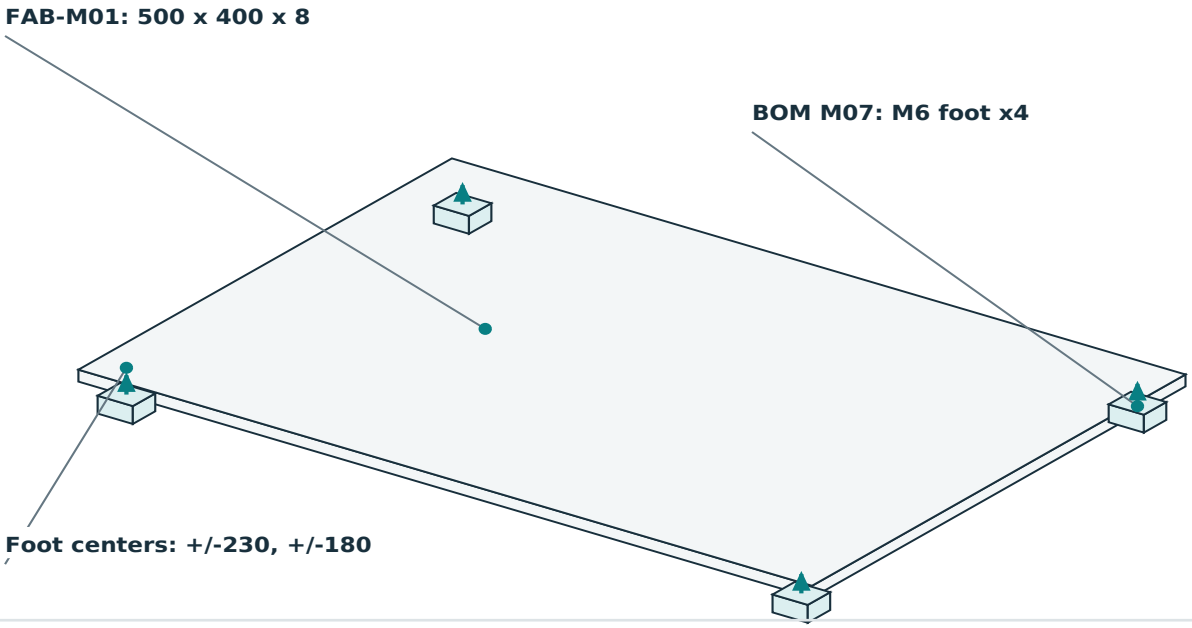
CHECK BEFORE CONTINUING

Noncritical stock dimensions within +/-0.5 mm; hole positions within +/-0.2 mm or the released tighter requirement. Bearing fits follow the bearing supplier.

Fabrication drawing rev: _____ Inspection initials: _____

Build the base and support structure

PARTS FAB-M01 x1; BOM M07 rubber feet x4; F03 M6 hardware; G01 guard-support material per released H3 drawing.
TOOLS Hex keys, square, straightedge, torque tool



1. Place the base on a protected surface. Fit one M6 foot at each corner hole with a washer and locking arrangement compatible with the delivered foot. Select bolt/stud length using the fastener worksheet.
2. Turn upright and level on a rigid bench. Keep X left/right, Y along the hinge, Z upward; negative Y is the operator/front side. Mark the centerline and the two support stations at Y=-95 and +95 mm.
3. Attach the released guard outriggers or support plinth. The guard is wider and deeper than the base; do not leave it supported only by its plastic panels. Exact supports are an unresolved design interface H3.
4. Trial-place the electrical enclosure outside the swept envelope. Leave access to the E-stop, reset, power plugs, service cover and front door.

CHECK BEFORE CONTINUING

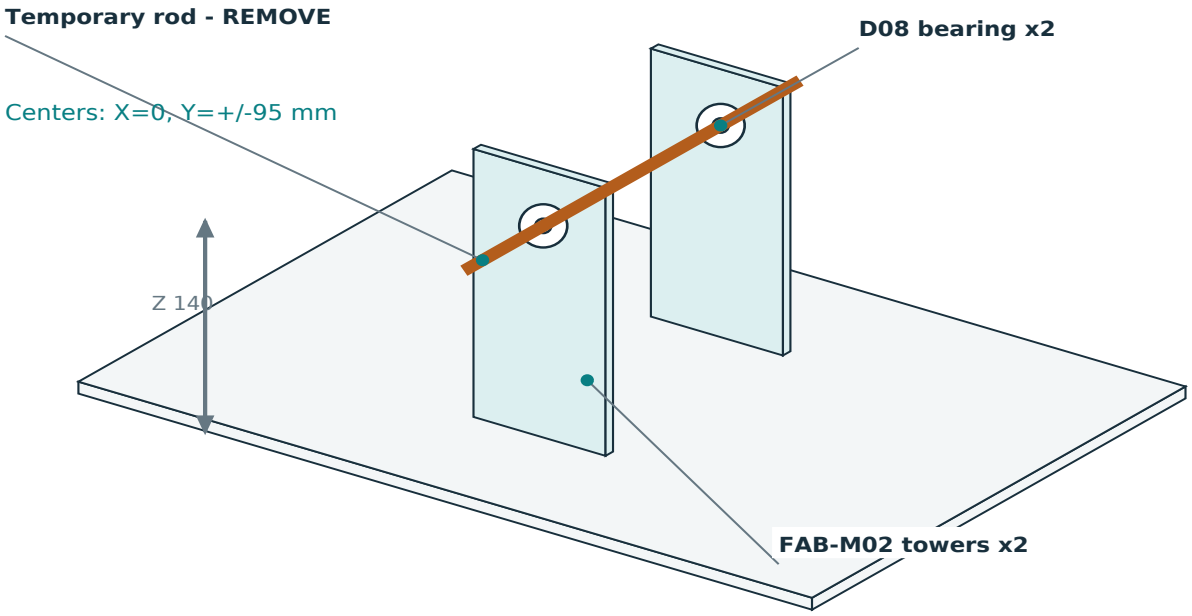
Base is stable with no rocking. Support diagonals differ by no more than 1 mm. Guard and enclosure loads have a positive load path to the bench.

Foot fixing torque/source: _____ Base level check: _____

Align the two bearing towers

PARTS FAB-M02 towers x2 with angle feet/adapters; BOM D08 bearings x2; F03 M6 fixings; F04 shims; temporary 6 mm alignment rod.

TOOLS Height gauge or caliper, indicator, hex keys



1. Bolt each tower foot loosely to the base. Place bearing centers at Y=-95 and +95 mm. Use the tower slots to bring both centers to X=0 and Z=140 mm above the base top.
2. Insert a straight temporary rod through both bearings. Adjust feet and bearing adapters until the rod slides freely without levering either tower. Measure center height and lateral alignment at both ends.
3. Tighten progressively, rechecking after each stage. Target bearing-axis mismatch no greater than 0.2 mm. Do not use tightening force to bend the alignment rod into position.
4. Remove the alignment rod and label it SETUP ONLY. It must never remain across the phone hinge space.

CHECK BEFORE CONTINUING

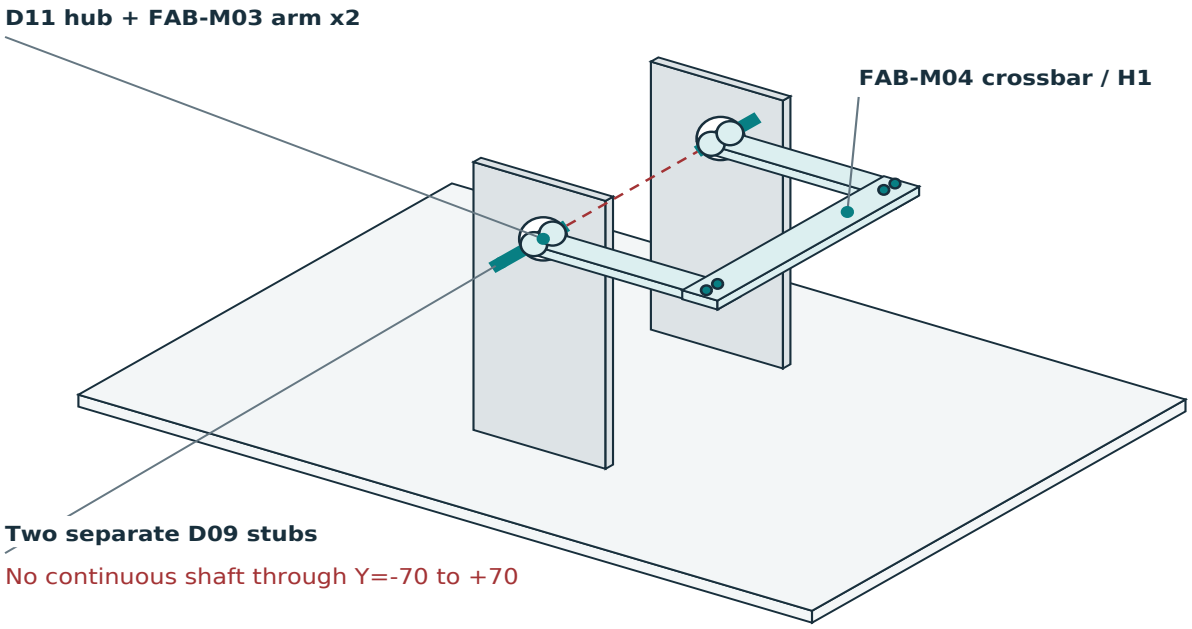
Alignment remains within target after tightening; rod removed. Mounting screws do not bottom or deform bearing housings.

Front X/Z: _____ Rear X/Z: _____ Rod removed: [ ]

Install stubs and the rotating yoke

PARTS BOM D09 shaft stock cut to accepted stack; D10 collars x4; D11 arm hubs x2; FAB-M03 arms x2, FAB-M04 crossbar x1; four M4 outer-joint fasteners.

TOOLS Hex keys, caliper, indicator, torque tool



1. Use two separate 6 mm stubs. The front stub projects toward the drive; the rear is an idler. Derive cut lengths from the delivered bearing, collar, hub and clutch engagements. The SCAD lengths are envelope references only.
2. Install a collar on each side of each bearing support, allowing free rotation without axial clamping of the bearing. Keep shaft ends and all drive hardware outside the phone keep-out region Y=-70 to +70 mm.
3. Fit the two split-clamp hubs and released radial arms. Join the outer crossbar at nominal X=115 mm using two M4 fasteners per joint. Close H1 before drilling; align the real hub centers with the shaft axes.
4. Rotate the empty yoke by hand through its planned arc. Recheck collar clearance, screw engagement and rigidity, then add witness marks.

CHECK BEFORE CONTINUING

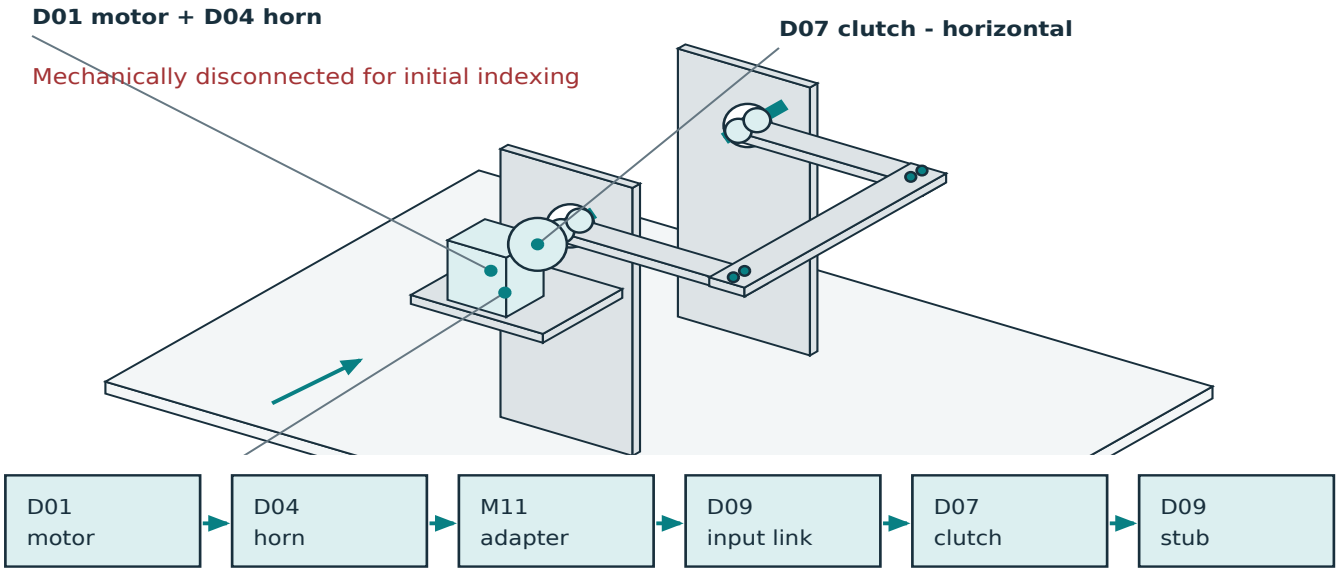
Empty hinge space; free motion without a tight spot; no axial escape. Both hub/arm joints follow a released drawing, not the old envelope model.

Stub lengths front/rear: _____ Yoke drawing rev: _____

Fit the motor and slip clutch

PARTS BOM D01 XM430-W350-T; D04 horn/bracket; D07 confirmed EAS12 clutch; D09 input link; D10 remaining collars as required; FAB-M09 shelf, FAB-M11 adapter.

TOOLS Indicator, hex keys, low-range torque tool



Axial assembly order; no shaft through phone

1. Install the horn with the supplied ROBOTIS screws and instructions. Transfer its actual bolt pattern to FAB-M11. Verify 6 mm bore fit, clamp retention, screw depth and clearance; never invent spline dimensions.
2. Attach the motor bracket to the slotted shelf outside the front tower. Arrange the axial chain: motor/horn, adapter, short input link, clutch, front stub, supported yoke.
3. With the clutch disconnected, align the drive axis to the supported stub; target no more than 0.1 mm offset. Connect without forcing the shafts sideways. Do not use the servo output bearing as the sole fixture support.
4. Keep the clutch horizontal. The EAS12-6MM-6MM is a requested configuration: verify supplier acceptance, shaft engagement, both-direction adjustment and mounting instructions before fitting. Leave mechanically uncoupled for electrical/indexing steps.

CHECK BEFORE CONTINUING

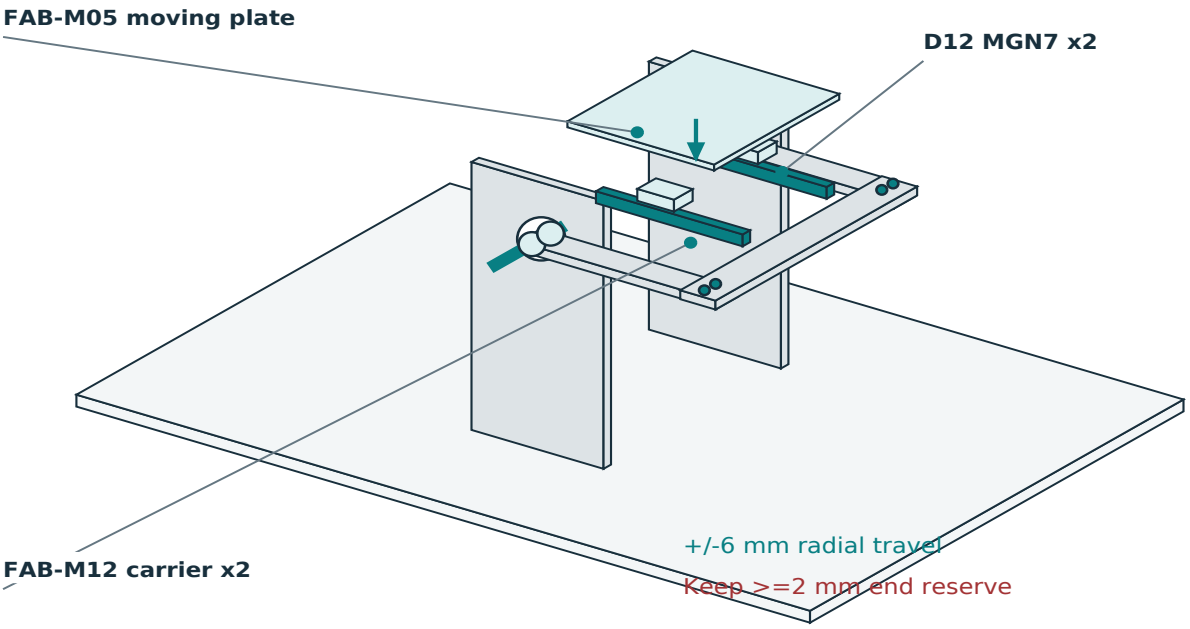
All clamp engagements measured and recorded. The adapter and clutch turn without binding; coupling remains open until step 23.

Clutch serial/configuration: _____ Alignment: _____ mm

Build the radial floating carriage

PARTS BOM D12 MGN7 rails/carriages x2; FAB-M12 carriers x2; FAB-M05 moving plate x1; vendor rail screws; F02 M4 plate hardware.

TOOLS Straightedge, caliper, hex keys



1. Fit the two guide carriers across the yoke. Orient both 100 mm rails radially, toward/away from the hinge in the moving half's plane. Use the delivered rail hole pattern and prescribed screws, not assumed M4 rail screws.
2. Mount one rail as the reference; leave the second slightly loose. Attach the plate to both carriages through its adjustment slots. Move the plate gently and tighten the second rail without introducing a tight spot.
3. Center the available float and fit positive end retention so no carriage can leave its rail. Use released retainers; screws in a rail hole are acceptable only if they cannot strike rolling elements or violate the rail instructions.
4. Check at least +/-6 mm of usable radial float. During the eventual phone sweep, preserve at least 2 mm travel reserve at each end; revise the mount if the hinge consumes that reserve.

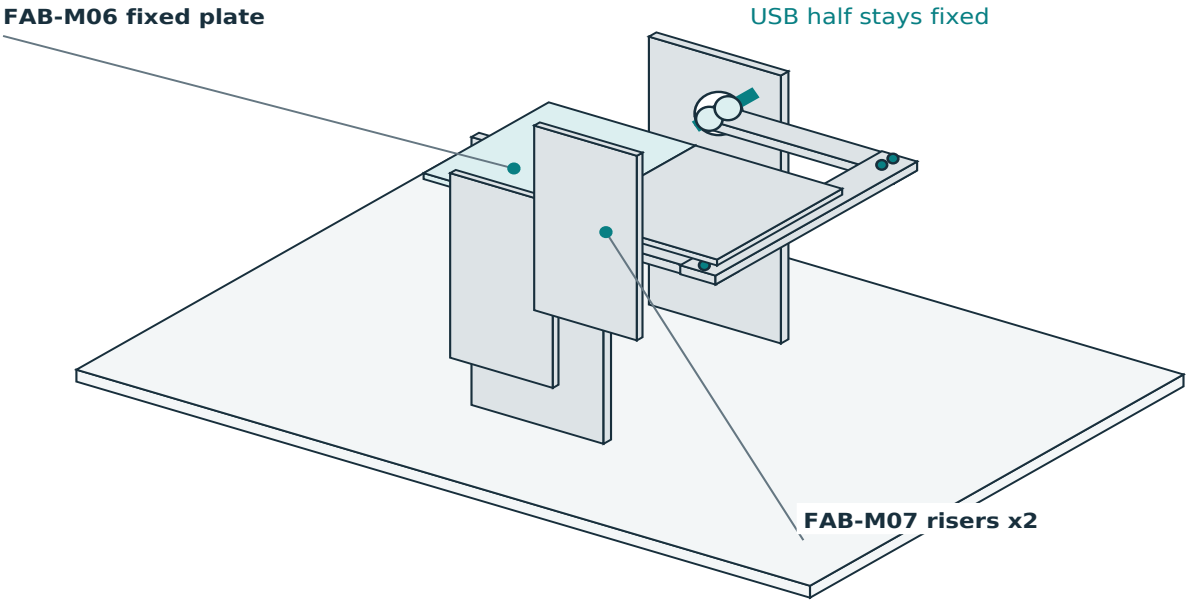
CHECK BEFORE CONTINUING

Plate moves freely by hand, without appreciable preload, binding or loose carriage retention. H2 carrier and retention drawing is released.

Measured radial stroke: _____ Rail fixing torque/source: _____

Install the fixed cradle

PARTS FAB-M06 fixed plate x1; FAB-M07 risers/feet x2; F02/F03 hardware; F04 shims; BOM T05 dummy.
TOOLS Caliper, straightedge, angle gauge, shims



1. Attach the two risers on the negative-X side of the hinge. Loosely mount the fixed plate using its four 4.5 x 20 mm slots. The USB-port half is stationary.
2. Use the dummy and measured back-surface offsets to shim the fixed backing plane relative to the moving cradle at the open position. Z=140 mm is the hinge-axis datum, not automatically the top of either pad.
3. Machine camera, button and connector relief from the actual measurements. Keep support on rigid exterior frame/back lands. Do not copy one phone half's relief onto the other.
4. Manually sweep the empty/dummy mechanism. Check the fixed plate and risers against the rotating yoke, moving plate and screw ends throughout the arc, then tighten and mark.

CHECK BEFORE CONTINUING

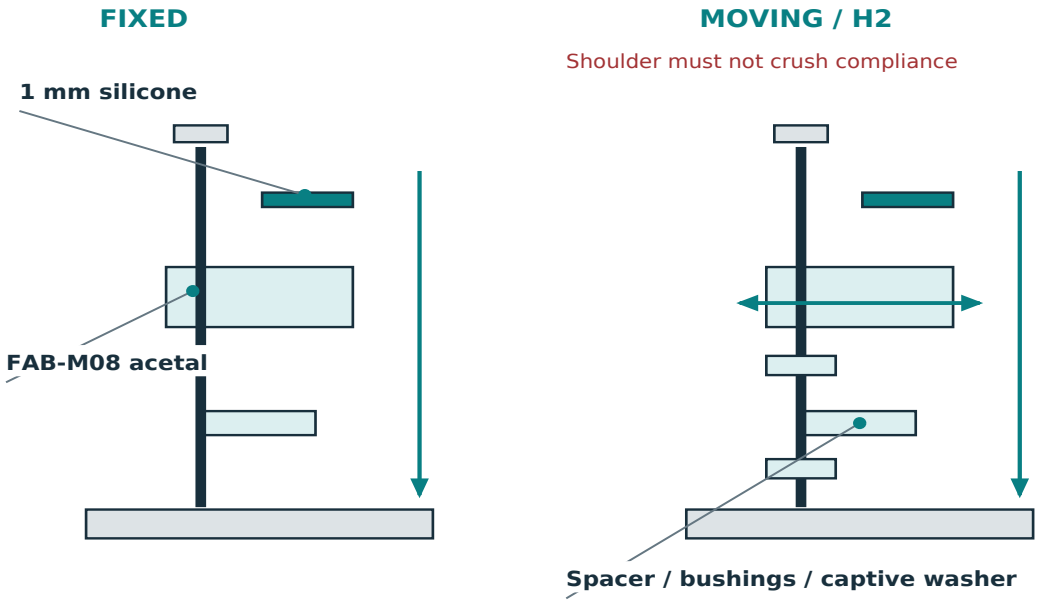
No interference at any angle; the plates do not force either leaf out of its natural plane. USB plug/strain relief space remains available.

Backing-plane offsets: _____ Minimum observed clearance: _____

Fit compliant shoes and pads

PARTS FAB-M08 acetal shoes x8 (BOM M03); BOM M04 1 mm silicone pad sheet; BOM M05 four shoulder screws, spacers, eight bushings and retainers.

TOOLS Caliper, knife, hex keys, force gauge



Exploded concept; H2 defines shoulder, spacer and retention dimensions.

1. Machine four shoe contact contours for each half from the actual frame map. Add 1 mm silicone pads, Shore A 30-40. Keep all shoes away from screen, hinge gap, buttons, lens glass and connectors.
2. Fix four shoes to the stationary plate. At the moving plate, assemble each of the four compliant mounts with the released shoulder-screw/spacer/bushing stack. The shoulder/spacer carries tightening load; do not crush the bushings solid.
3. Set approximately +/-2 mm tangential compliance with positive capture. Spacer length, shoulder length and washer diameters are H2 release items. A conceptual exploded stack is shown; it is not a dimensioned production joint.
4. Fit the dummy; check a 10 N tangential retention pull without slip. Later repeat phone fit gently within approved contact forces. If retention requires screen loading or excessive clamp pressure, redesign the shoes.

CHECK BEFORE CONTINUING

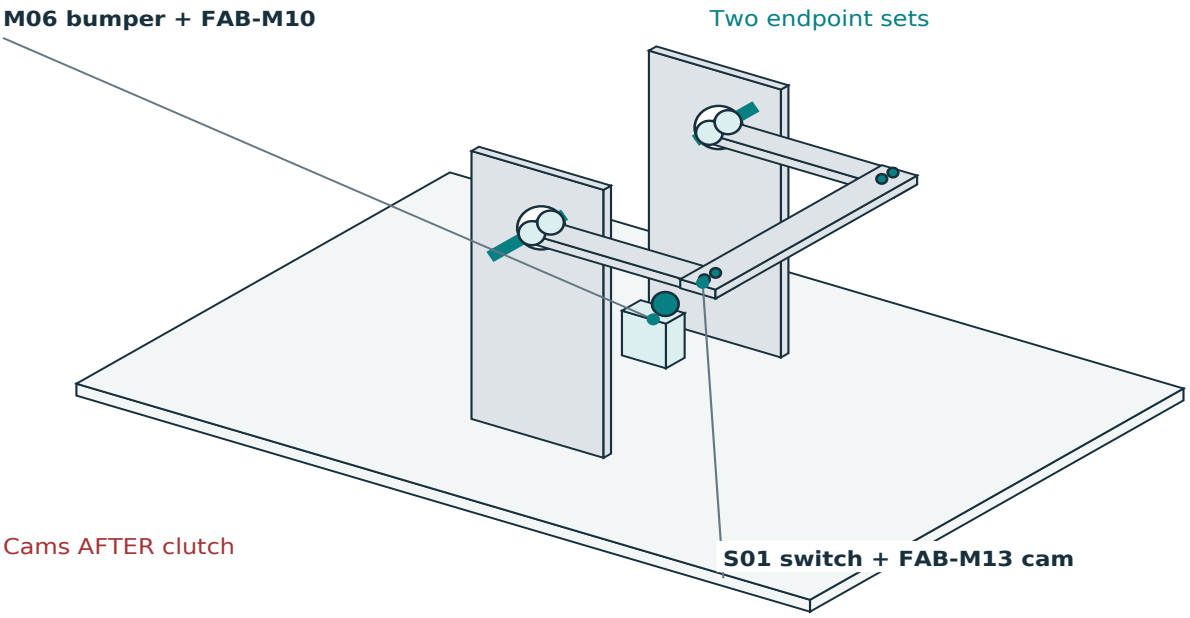
All eight pads secure; moving mounts float and remain captive. Clamp witness marks can be seen with the guard shut.

Shoe drawing rev: _____ Dummy retention result: _____

Set stops and endpoint switches

PARTS FAB-M10 brackets x4; BOM M06 bumper stops x2; S01 D2F-01L switches x2; FAB-M13 cams x2; F01/F02 mounting hardware.

TOOLS Angle gauge, multimeter, hex keys



1. Fit one mechanical stop at each end of yoke travel. Both act on fixture structure, never on the phone. With a dummy, set an initial 5-175 degree operating span; place stops about 1 degree beyond endpoints only if that remains within safe free travel.
2. Mount switches independently of the stops. Use output-side cams on the yoke after the clutch; motor rotation alone must not prove that the phone moved.
3. Adjust each cam to actuate only close to its endpoint, leaving switch overtravel within its datasheet. The closed switch must release before the open switch can operate, and vice versa.
4. Lock all adjustments, add witness marks and measure physical stop/switch angles. Re-set for the real specimen before any full-range operation.

CHECK BEFORE CONTINUING

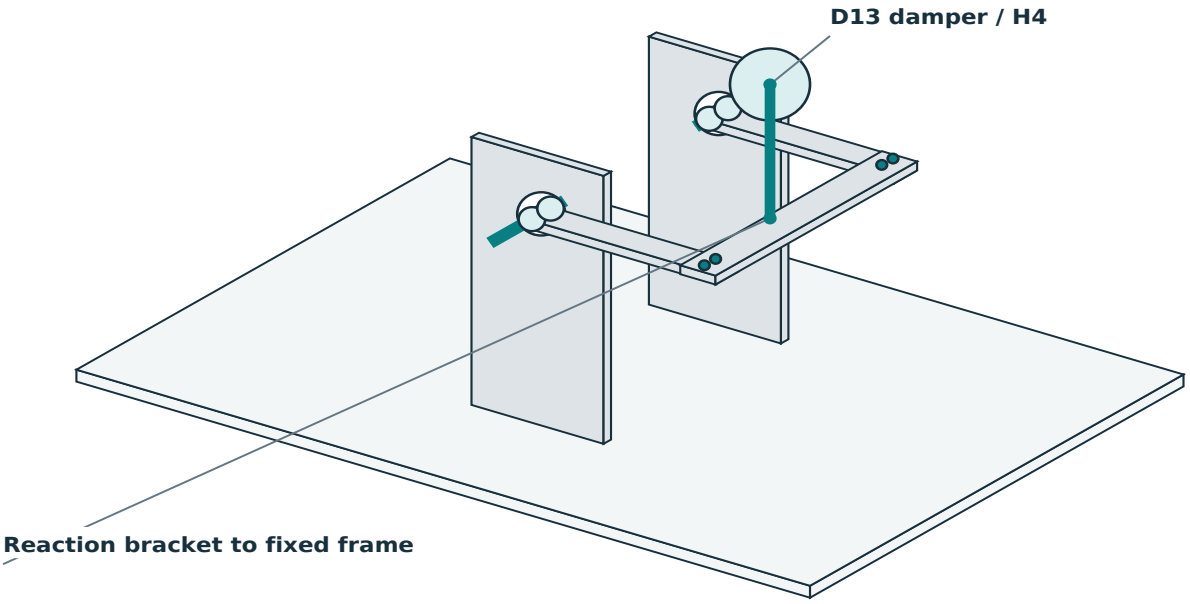
Neither switch is a mechanical stop. Both cannot be active together. The yoke cannot drive the specimen into negative opening or overextension.

Closed switch/stop: _____ Open switch/stop: _____ degrees

Install and qualify the damper

PARTS BOM D13 bidirectional damper; released H4 reaction bracket/coupling; dummy; fasteners specified on accepted damper drawing.

TOOLS Force gauge, angle gauge, video/time reference



1. Select the damper from measured moving mass/inertia and expected gravity settling. The BOM contains an allowance, not a validated make/model or mounting pattern. Release H4 and procure its bracket/adapter before proceeding.
2. Attach the damper reaction member to a stationary support and its moving member to the yoke output side. Keep it clear of the phone and rails; confirm it resists both directions without becoming a hard stop.
3. With the motor mechanically uncoupled and a restrained dummy, release at several angles. Record gravity movement and time until stationary; keep hands outside the swept area.
4. Check added friction against the available clutch/current range. Excessive drag requires a different damper or counterbalance design, not a higher unqualified motor limit.

CHECK BEFORE CONTINUING

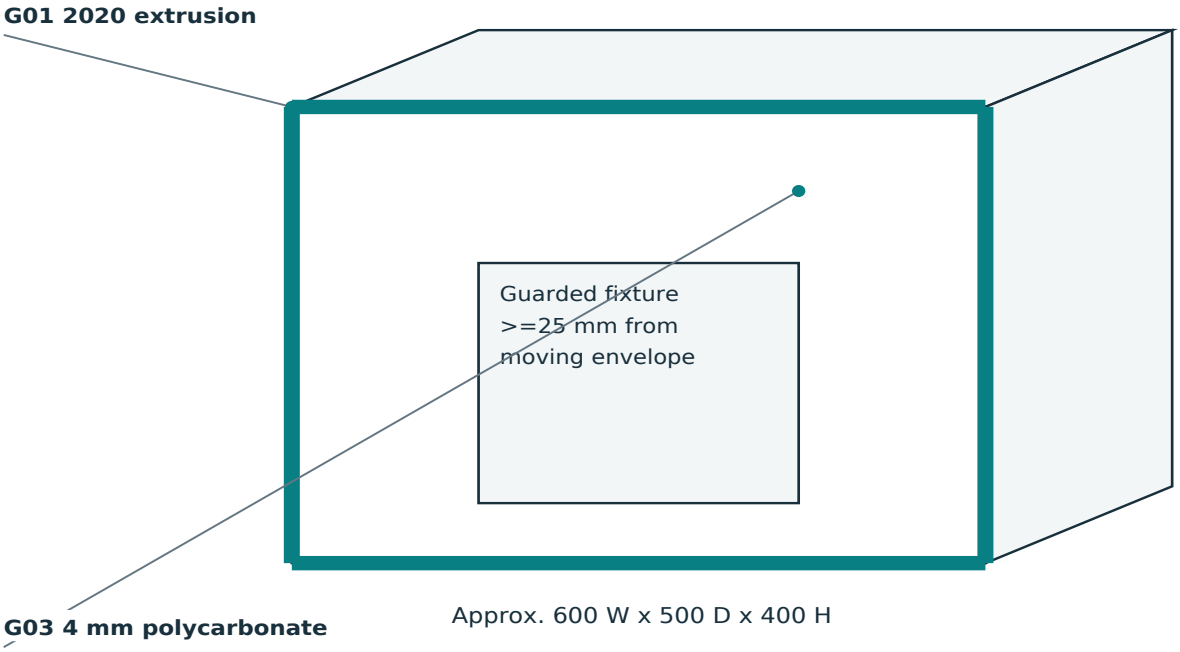
Motion under power loss is measured. The damper is not credited as a safety-rated brake; guard access depends on the later stopping test.

Damper model/rev: _____ Worst settling angle/time: _____

Assemble the guard frame and panels

PARTS BOM G01 2020 frame; G02 brackets x24; G03 five 4 mm panels/FAB-M14; G04 clips/fasteners; released H3 supports.

TOOLS Square, tape, hex keys, plastic drill/deburr tools



1. Dry-build the frame using matching slot-series parts. Stock schedule: four 560, four 460 and four 360 mm members. Confirm the chosen joint arrangement before cutting: these lengths alone do not define every overlap or final opening.
2. Mount the frame to the released support structure. Nominal external envelope is 600 x 500 x 400 mm. Sweep the complete mechanism by hand; maintain at least 25 mm to every moving item, including cable loops.
3. Fit rear, sides and top panels with captive clips and thermal-expansion clearance per the panel/fastener supplier. Nominal panels: 560 x 360 x2, 460 x 360 x2, 560 x 460 x1; trial-fit before final drilling.
4. Reserve the front panel for the door. Keep edges covered or rounded; assess all seams, ventilation slots and cable entries for reach-through.

CHECK BEFORE CONTINUING

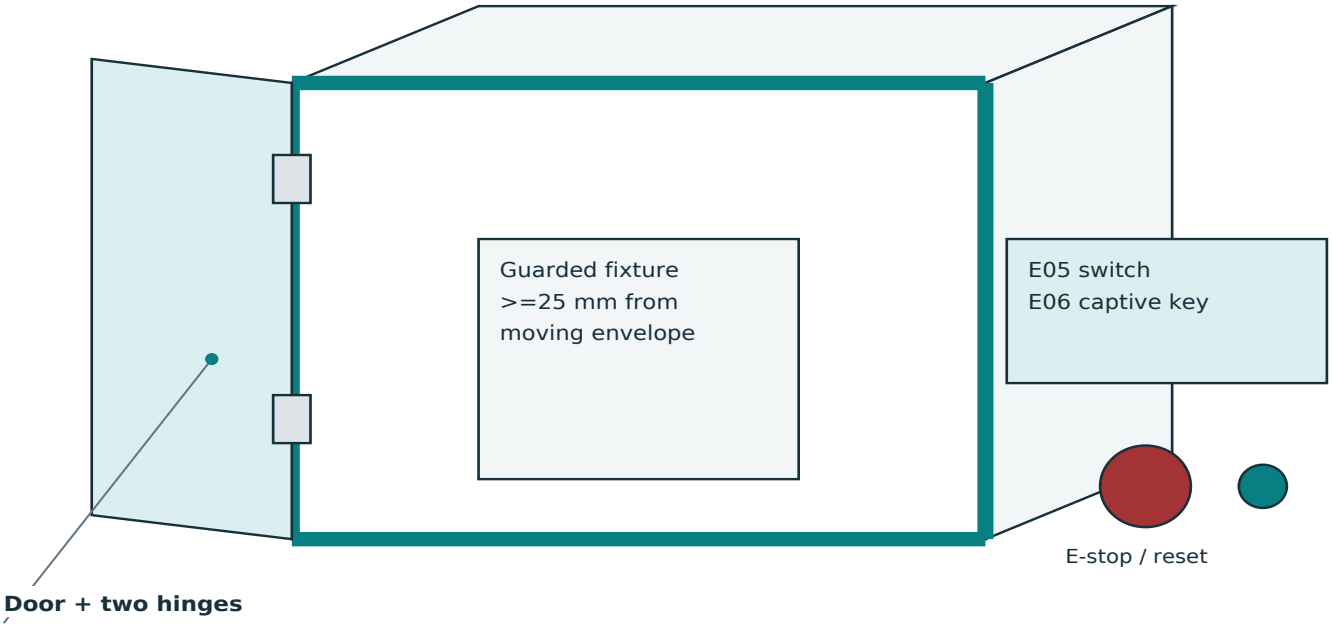
Frame square within 1 mm diagonal difference, securely supported; panels do not carry frame loads. 25 mm motion clearance is not a finger-access safety rating.

H3 drawing rev: _____ Smallest swept clearance: _____ mm

Hang the door and fit controls

PARTS BOM G04 hinges x2, handle/latch; E05 D4NS-1BF; E06 D4DS-K2 key; E04 two-NC E-stop; E07 NO reset; mounting panel/brackets per H3.

TOOLS Drill, hex keys, screwdriver, multimeter



1. Hang the front panel on two hinges. Install a positive door stop and latch; the switch body/key must not serve as the door's mechanical stop. Confirm smooth closing without bending the key.
2. Secure the tongue key captively to the door, and the switch to the fixed frame. Verify the delivered D4NS entry direction and operating stroke against its drawing. Use its specified gland and tightening values. [R4]
3. Mount the red/yellow E-stop and labeled physical reset outside the guard, reachable from the operator position with a clear view of the fixture. Record both E-stop NC contact labels before wiring.
4. With power absent, verify both guard NC paths open when the door opens and both E-stop NC paths open when pressed. Final access timing is tested in step 24.

CHECK BEFORE CONTINUING

Closing the door or releasing the E-stop cannot start motion. If coast-down allows access to moving parts, a validated guard-locking redesign is required.

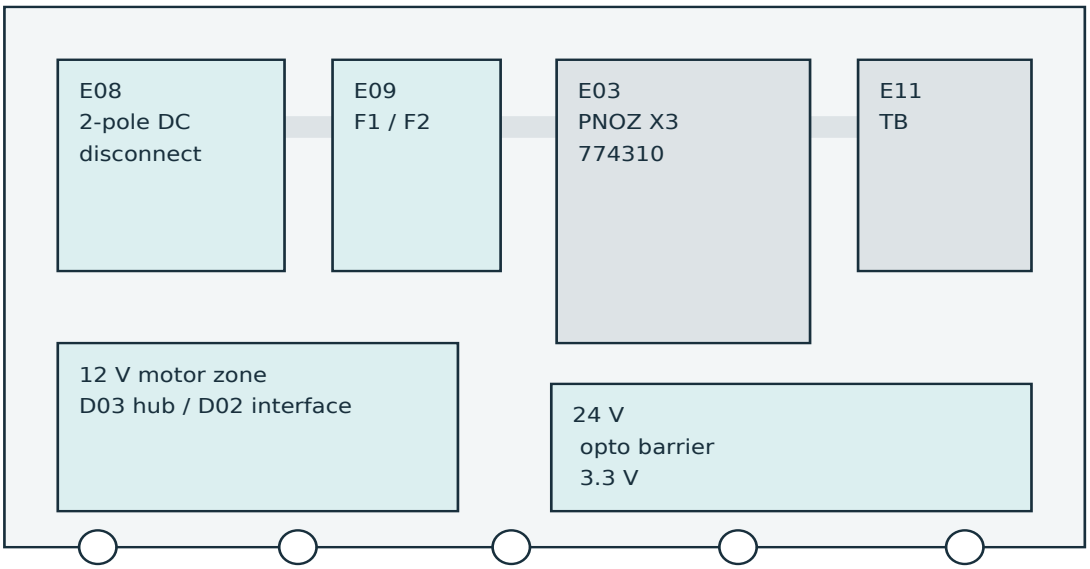
Switch/key alignment: _____ Contact IDs verified: _____

Mount the electrical enclosure

PARTS BOM E10 enclosure; E11 DIN/terminals; E03 PNOZ X3 774310; E08 DC disconnect; E09 F1/F2; W03 glands/ferrules; D02/D03 interface/hub.

TOOLS Screwdriver, crimper, drill, label printer

Enclosure layout concept - verify delivered clearances



1. Mount the nominal 300 x 250 x 150 mm covered enclosure outside the motion envelope. Arrange DIN rail, relay, fuse holders and labeled terminals with wiring space and access to every screw.
2. Define three labeled wiring zones: switched 12 V motor, 24 V safety, and 3.3 V Pi signal. Mount the optocoupler board on insulated standoffs with a physical gap between its 24 V and Pi sides.
3. Provide strain-relieved entries for two external SELV bricks, motor/USB, door/E-stop/reset, and sensors. Use external enclosed supplies; no exposed mains terminations are part of this design.
4. Mark the 2-pole disconnect as MOTOR + SAFETY DC ISOLATOR: one pole interrupts +12 V, the other +24 V. The separately powered Pi remains live; isolate all three supplies before internal service.

CHECK BEFORE CONTINUING

Cover can close without crushing wires; service isolation is clear. All unused relay terminals insulated; no loose conductive debris.

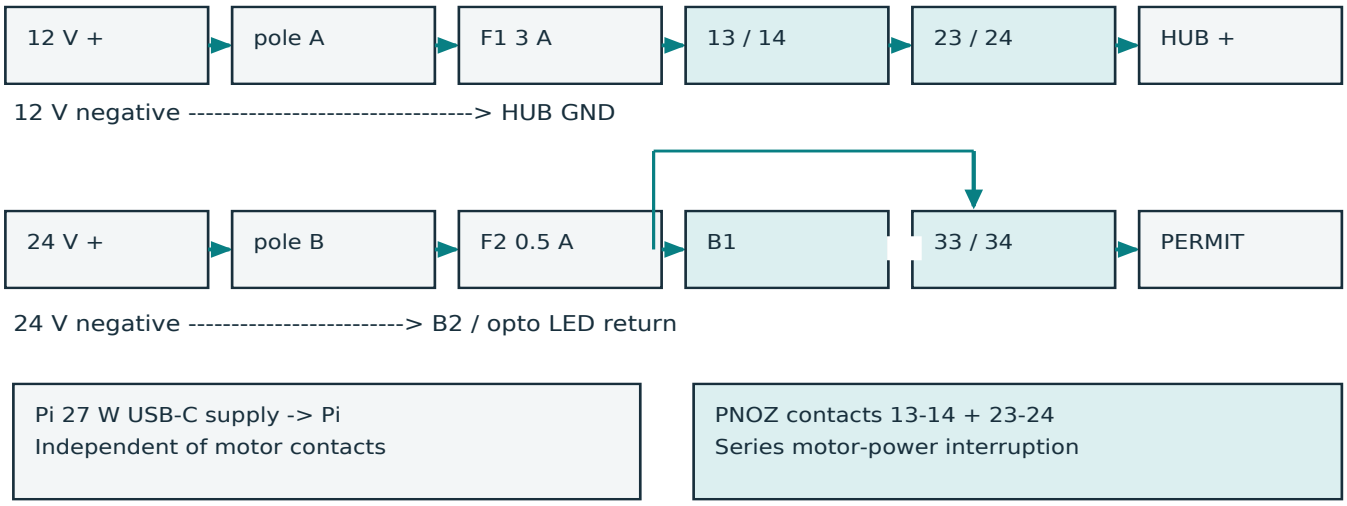
Panel drawing rev: _____ Terminal/fuse labels checked: _____

Wire motor and relay power

PARTS BOM E01 12 V/5 A, E02 24 V/1 A supplies; E08 disconnect; F1 3 A fast DC, F2 0.5 A DC; E03 relay; D03 hub; W01/W02 wire.

TOOLS Crimper, insulated screwdriver, multimeter

Power paths / keyed hub pinout checked on delivered part



1. With bricks unplugged, wire +12 V through disconnect pole A, F1 and relay terminal 13. Jumper 14 to 23; connect 24 to the hub + input. Connect 12 V negative directly to hub motor ground. Use 18 AWG for this branch.
2. Wire +24 V through disconnect pole B and F2 to relay B1. Wire 24 V negative to B2. B1/B2 are the DC supply terminals for this relay variant; do not substitute A1/A2. [R1]
3. Branch fused +24 V to relay 33. Relay 34 feeds the isolated permit board in step 17. The two series motor contacts remove +12 V; status contact 33-34 only reports relay state.
4. Keep the servo unplugged. Verify connector pin numbers/polarity at the delivered hub and confirm there is no alternate unswitched supply path. Record actual fuse part numbers and check contact/load/inrush coordination.

CHECK BEFORE CONTINUING

Positive motor supply passes through both 13-14 and 23-24. No 24 V conductor reaches the Pi or motor positive input.

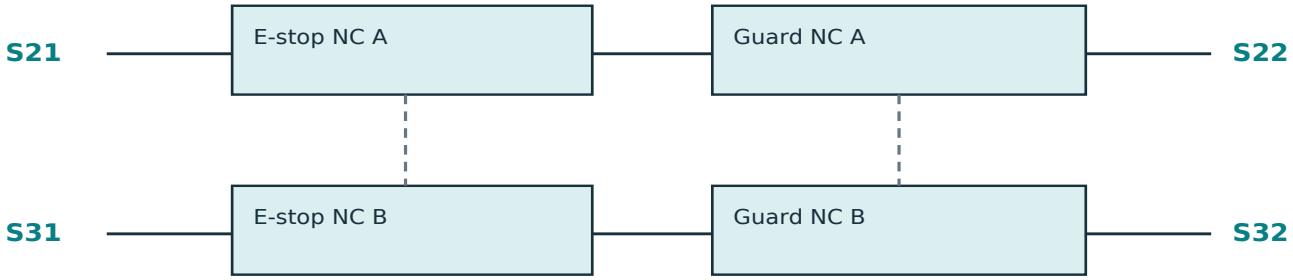
F1/F2 part numbers: _____ Continuity/polarity initials: _____

Wire the two safety channels

PARTS BOM E03 PNOZ X3; E04 E-stop contacts A/B; E05 guard NC contacts A/B; E07 NO reset; W02 22 AWG safety wire.

TOOLS Multimeter, ferrule crimper, insulated screwdriver

R1 pp.12-13 + project series-device adaptation / H5 review



NC by function: both open on demand. Verify device terminal IDs.

S11 ---- S12 Link for this circuit	S33 -- NO reset -- S34 Monitored physical start	S13 / S14 NO LINK

1. Use the two-channel cross-short-detection input arrangement in Pilz 20547-EN-10 page 12, with the monitored-start arrangement on page 13. Retain the S11-S12 link for this selected circuit. Verify against the delivered relay revision. [R1]
2. Channel A: S21 -> E-stop NC-A -> guard NC-A -> S22. Channel B: S31 -> E-stop NC-B -> guard NC-B -> S32. Keep the channels separately labeled; verify actual switch contact markings with the meter.
3. Wire the momentary NO reset between S33 and S34. Leave S13/S14 unlinked. Do not apply external 24 V to these input terminals or substitute an API reset for the physical pushbutton.
4. Record this as a project adaptation using series-connected safeguards, not a certified whole-machine circuit. Have the electrical reviewer assess fault masking, short detection and reset behavior; test each device and channel separately.

CHECK BEFORE CONTINUING

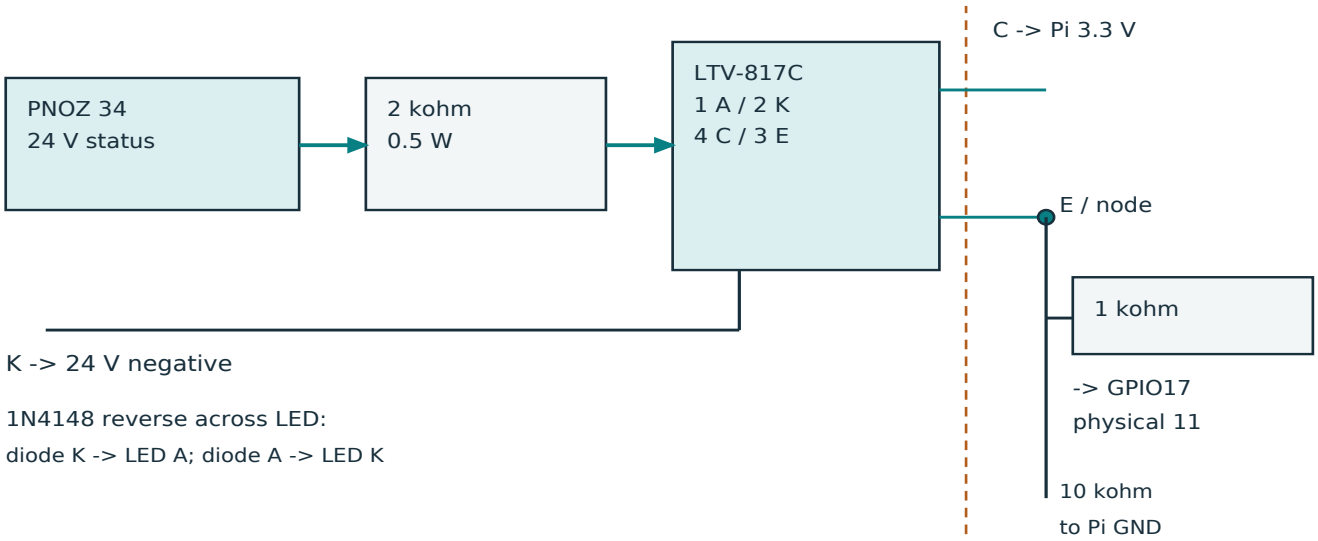
Both devices interrupt both channels. No automatic-start link. H5 electrical review and a point-to-point check are complete before step 21.

Relay manual revision: _____ Electrical reviewer: _____

Build the isolated permit input

PARTS BOM S03: LTV-817C; 2 kohm 0.5 W; 1N4148; one 10 kohm and one 1 kohm; insulated board/terminals; W04 breakout.

TOOLS Soldering tools, magnifier, multimeter



R8 pin functions shown; verify pin-1 mark on delivered package.

1. On the 24 V side, connect relay 34 through the 2 kohm resistor to LED anode; LED cathode returns to 24 V negative. Put the 1N4148 across the LED in reverse: diode cathode at LED anode, diode anode at LED cathode.
2. On the Pi side, connect phototransistor collector to 3.3 V. Connect emitter to the PERMIT node, 10 kohm from node to Pi ground, then 1 kohm from node to GPIO17 (physical pin 11).
3. For the LTV-817 four-pin package, the manufacturer identifies 1=anode, 2=cathode, 3=emitter and 4=collector. Locate the pin-1 mark and verify the actual package drawing before soldering; do not mirror the bottom view. Attach the checked pin map to H5. [R8]
4. Keep the two domains separated without a ground jumper. At 24 V the LED current is approximately 11 mA and resistor loss about 0.26 W. Confirm the actual resistor rating, CTR and temperature before closing the board cover.

CHECK BEFORE CONTINUING

With the Pi signal pin disconnected, powered node is >2.0 V and <=3.3 V when permitted, <0.8 V when off. No 24 V path to the Pi side.

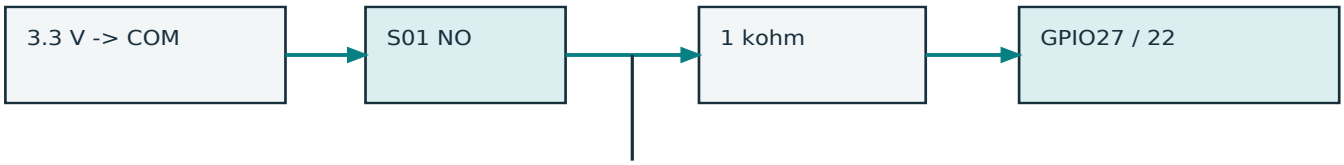
Optocoupler pin map/rev: _____ Node high/low: _____ V

Wire endpoints and temperature probes

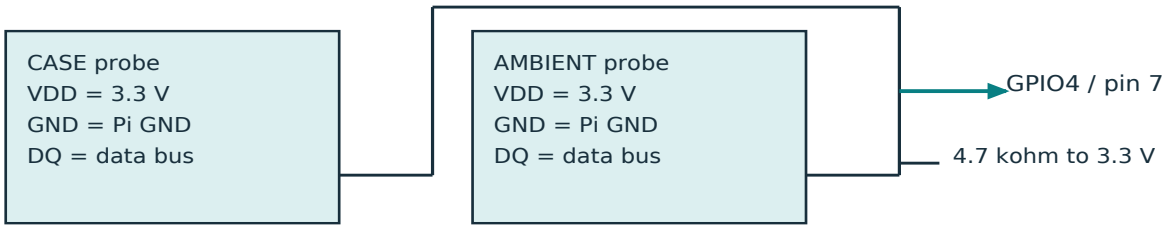
PARTS BOM S01 switches x2; S02 DS18B20 probes x2; remaining S03 resistors: 2 x 10 kohm, 2 x 1 kohm, 1 x 4.7 kohm; S04 attachments; W04 breakout.

TOOLS Multimeter, screwdriver, labels, thermal tape

Repeat endpoint circuit x2



Two powered DS18B20 probes / one shared pullup



1. For each endpoint, connect switch COM to Pi 3.3 V and NO to its signal node; add 10 kohm node-to-ground and 1 kohm node-to-GPIO. OPEN uses GPIO27/pin 13; CLOSED uses GPIO22/pin 15. Insulate unused NC contacts.
2. Power both DS18B20 probes from 3.3 V and Pi ground. Join both data wires to GPIO4/pin 7 and use one shared 4.7 kohm pullup to 3.3 V. Use powered three-wire mode. Identify probe wire functions from its supplier, not cable color. [R5]
3. Attach the case probe to the stationary exterior frame/back with removable thermal tape. Place the ambient probe nearby in shaded free air. Keep both clear of the hinge, lens, flexible display and moving clamp.
4. Label both probe serials after software identification. Confirm wiring includes a service loop and fixed strain relief without a loop entering the swept envelope.

CHECK BEFORE CONTINUING

Broken endpoint wire reads low. Each switch is high only at its own endpoint. Case probe stays attached throughout a manual sweep.

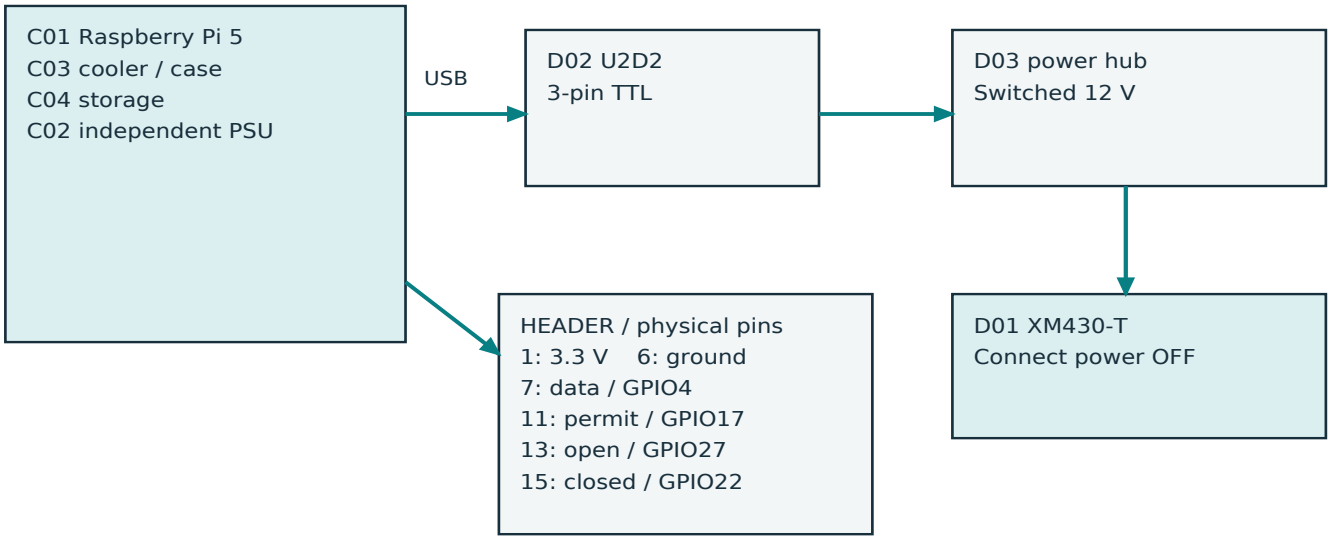
Case serial: _____ Ambient serial: _____ Wiring checked: _____

Install the Pi and finish the harness

PARTS BOM C01-C05 Pi/cooler/card/supply/network; D02 U2D2, D03 hub, D05 USB, D06 TTL cables x2; W03 strain relief; W04 breakout.

TOOLS Screwdriver, multimeter, cable labels

Identify pin 1 from board markings; avoid mirrored connector views.



1. Install the active cooler and storage, then secure the Pi case outside the motion envelope with ventilation clear. Feed it only from its independent official USB-C supply.
2. With all power off, connect the keyed breakout and verify physical pin 1 orientation against the Pi board. Used pins: 1=3.3 V, 6=ground, 7=GPIO4, 11=GPIO17, 13=GPIO27, 15=GPIO22. Do not use 5 V pins for these circuits. [R6]
3. Connect Pi USB to U2D2, and its 3-pin TTL port to the power hub and XM430-T cable path. U2D2 does not power the motor; verify the keyed cable mapping and leave the motor end disconnected for now. [R3]
4. Bundle motor, safety and signal wiring separately. Label both ends; clamp the fixed side of every service loop. Keep cables away from collar screws, clutch, rail ends and stop bumpers.

CHECK BEFORE CONTINUING

No connector can be pulled into motion. Pi remains powered independently of the motor contacts. Do not hot-plug DYNAMIXEL cables. [R3]

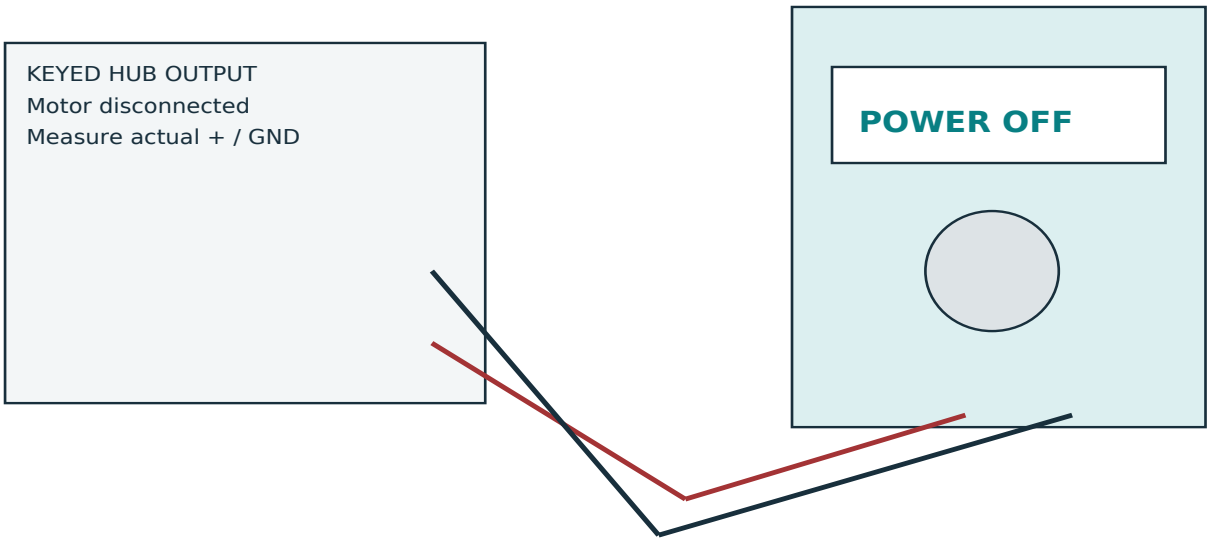
Header orientation checked: _____ USB/TTL continuity: _____

Inspect with every supply isolated

PARTS Assembled bench; BOM T03 meter; wire schedule and released H1-H5 drawings.

TOOLS Multimeter, torque tool, inspection light

Voltage mode only on energized circuits



Continuity mode only after zero-voltage verification

1. Unplug 12 V, 24 V and Pi supplies. Verify zero voltage before using continuity mode. Keep servo and Pi GPIO connector disconnected while checking the harness.
2. Check each wire against the wire schedule, including contact pairs, fuses, disconnect poles, negative returns and signal series resistors. Inspect ferrules for loose strands and tug each termination gently.
3. Verify there is no intentional connection between the isolated 24 V circuit and Pi signal ground. Account for any ground connection through USB on the 12 V motor side; do not infer whole-system galvanic isolation.
4. Inspect every fastener, collar, pad, cover and cable gland. Record measured fastener engagement/torque from approved sources. Close the electrical enclosure before energizing.

CHECK BEFORE CONTINUING

Signed point-to-point inspection; no direct 12/24 V path to a Pi GPIO. All H1-H5 release items closed for the build phase being tested.

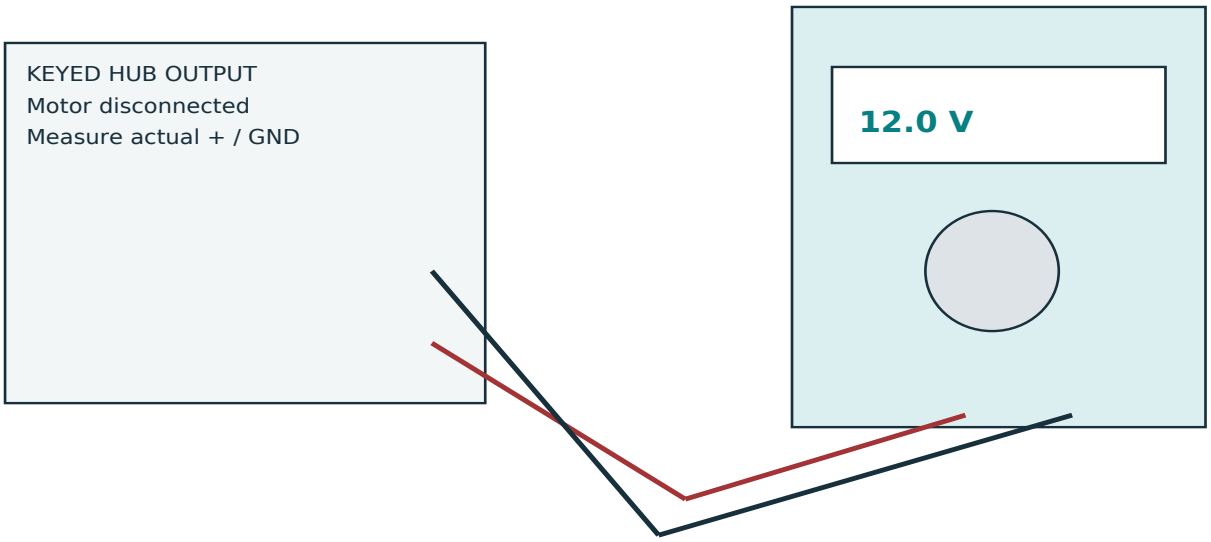
Electrical inspector/date: _____ Mechanical inspector/date: _____

Prove power removal with no motor

PARTS Servo disconnected; closed enclosure; meter at keyed motor-hub output; completed safety circuits.

TOOLS Multimeter, test record, second observer

Voltage mode only on energized circuits



Guard or E-stop open -> motor power absent

1. Apply power to the supplies with the servo unplugged. Verify relay supply polarity and voltage. With an open guard or pressed E-stop, motor-hub voltage must be near zero and permit low; investigate residual or alternate feed paths.
2. Close guard and release E-stop: motor supply must remain off until a valid physical reset. Reset should restore nominal 12 V at the hub and a valid permit level; it must not issue a motion command.
3. Operate E-stop and guard independently, then test each input channel by the approved electrical procedure. Verify motor voltage disappears. Check held-reset and power-restoration behavior; neither may bypass the intended monitored reset.
4. Isolate all supplies after testing. Only then connect the Pi signal harness and motor cable. Keep the drive mechanically uncoupled and the phone absent.

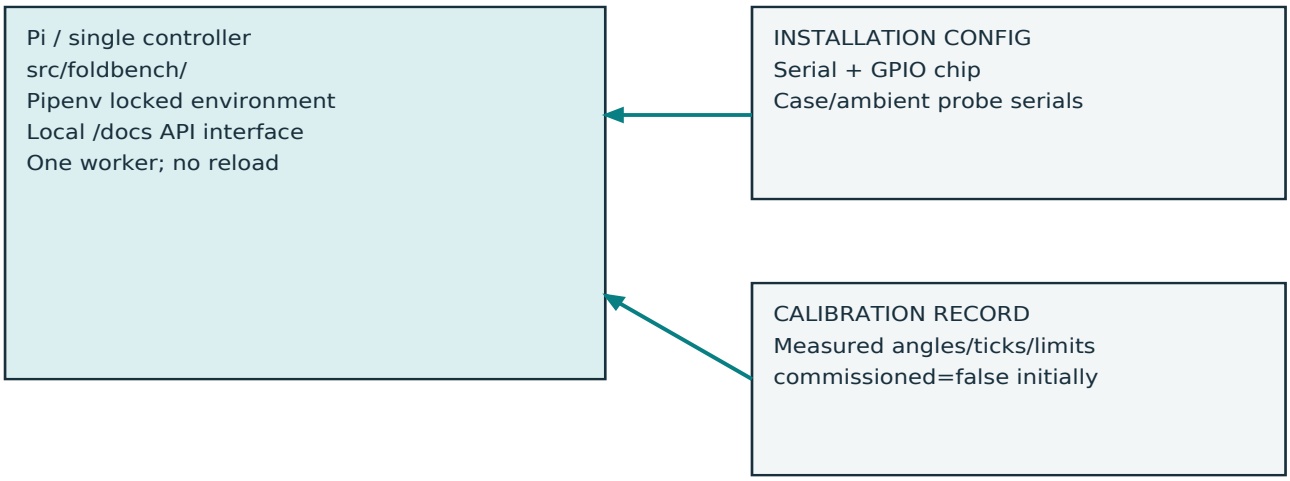
CHECK BEFORE CONTINUING

All rows of the static power-removal record pass. Failure means stop and correct wiring; a software status bit is not proof of disconnected motor power.

Hub enabled/off voltage: _____ Reset/E-stop/guard results: _____

Install and verify the controller

PARTS BOM SW01 source repository; Pi OS 64-bit; Python 3.12; Pipenv; deploy/ service and environment examples.
TOOLS Network terminal, repository README, commissioning log



Simulation -> static checks -> dummy authorization -> phone trial

1. Install the repository at /opt/foldbench and create its in-project virtual environment with Pipenv. Use the locked dependencies. First run simulation and confirm /health, authenticated /v1/status and the interactive /docs page.
2. Create a dedicated foldbench service account with dialout/gpio access and /var/lib/foldbench ownership. Configure the supplied single-worker systemd unit. Keep it stopped during probe/index work; do not run a second controller.
3. Set a random API key, persistent serial path, real GPIO chip path, data path and calibration path in the protected environment file. Enable the w1-gpio overlay for BCM4 using the OS instructions; reboot and identify both probe serials by warming one gently.
4. Check header-chip identity and Linux GPIO v1 ABI availability. Start with commissioned=false; use hardware mode only after verifying settings. Observe each input/probe; an unsupported kernel or missing probe is a stop condition, not permission to bypass a check.

CHECK BEFORE CONTINUING

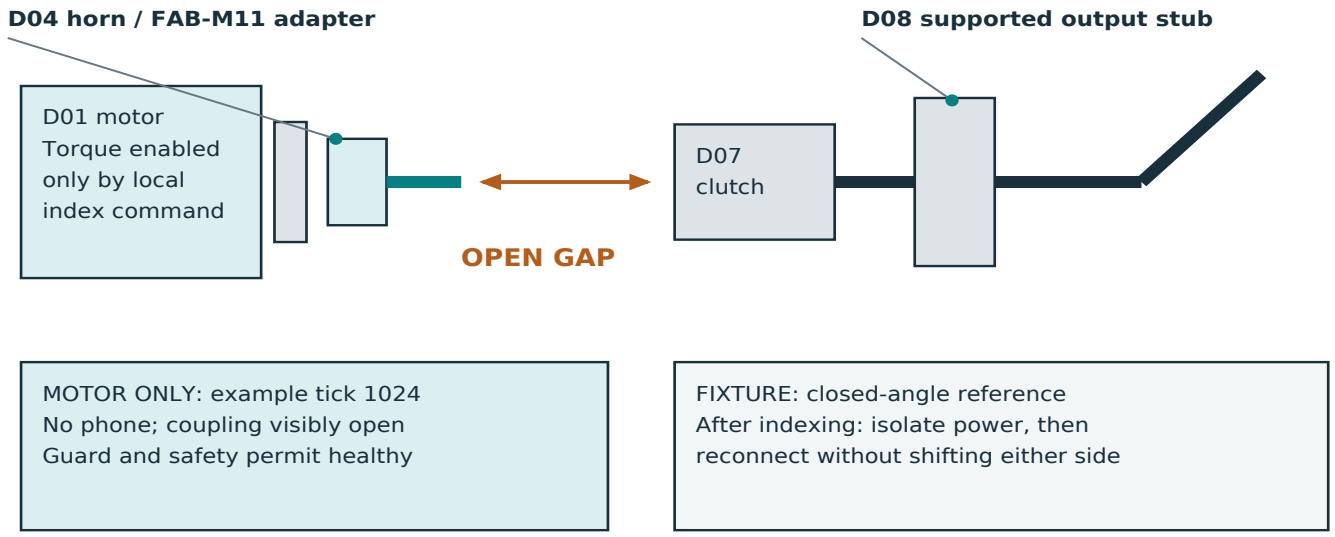
Simulation API works; serial/GPIO/probe identities are recorded. Hardware remains unable to command runs with the example uncommissioned calibration.

Git commit/OS/kernel: _____ Serial path/GPIO chip: _____

Index the uncoupled motor and align ticks

PARTS Uncoupled XM430-W350-T; empty fixture; probe utility; angle gauge; clutch connection still open.
TOOLS Terminal, angle gauge, witness marks

INDEXING SETUP: MOTOR DISCONNECTED FROM FIXTURE



1. Stop the service. Confirm the phone is absent and the motor is mechanically disconnected from the fixture. With guard/permit healthy, run the read-only probe and verify model 1020, firmware >=38, ID 1 and 57600 baud. Use vendor recovery for other ID/baud settings. [R2]
2. Only in that uncoupled condition, use the local probe indexing option shown in the software appendix to select a closed tick (example 1024). The command briefly enables low-current motion; --uncoupled is your declaration, not a coupling sensor.
3. Disable/isolate drive power, set the yoke at the measured closed endpoint, and reconnect the clutch without moving either side. Record closed angle/tick and provisional open tick = closed tick + angle span x 4096/360.
4. Verify positive ticks open the mechanism. For 5-175 degrees, 1024/2958 is only a starting example. Keep both ticks in 256-3839, their difference in 1500-2100 and angular agreement within 2 degrees as required by this software.

CHECK BEFORE CONTINUING

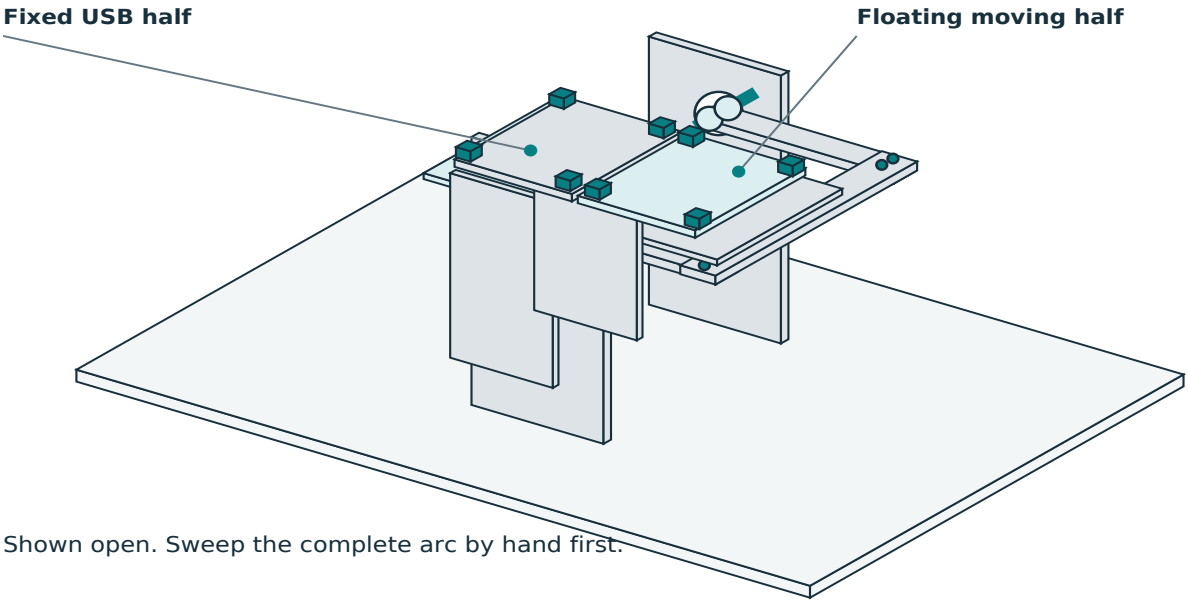
No coupled indexing command used. Independent angle measurements and switches agree with the recorded stroke; direction is correct before cycling.

Closed angle/tick: _____ Open angle/tick: _____

Qualify empty and dummy motion

PARTS Empty fixture, then BOM T05 dummy; completed guard; qualified clutch/damper; temporary EMPTY-FIXTURE calibration record.

TOOLS Angle/force gauges, independent tally/video, log



Shown open. Sweep the complete arc by hand first.

1. After static checks, create a signed dummy-only calibration with 5 degrees/s and the lowest current that moves the empty fixture within the installation ceiling. Only that supervised test record may set commissioned=true; it is not a phone release.
2. With guard closed, perform one endpoint jog at a time. Check direction, switch arrival and physical clearance. Operate E-stop and guard during motion; measure worst stopping/settling time and distance, then prove access is prevented until motion stops.
3. Test USB disconnection, Pi process/power loss, each endpoint/probe disconnection and a gentle instrumented dummy jam. Require faults, no false cycle increments and no automatic resumption. A watchdog fault requires inspection and motor power cycling.
4. Complete 100 dummy cycles, inspect, then 1,000. Record temperatures, cable wear, rail condition, witness marks, count agreement and clutch repeatability. Stop on any drift or wear.

CHECK BEFORE CONTINUING

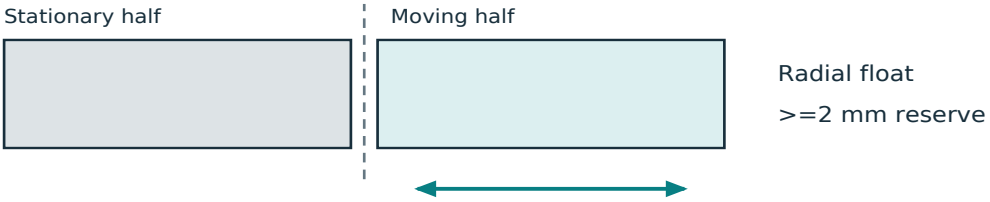
All fault injections and dummy runs recorded. If guard access precedes motion cessation, redesign guarding/locking before real-phone trials.

Dummy record ID: _____ Worst stop time/distance: _____

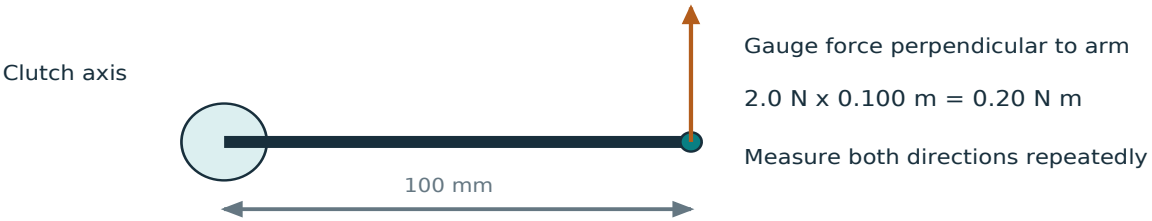
Fit the phone and measure protection

PARTS Actual phone; qualified shoes; BOM T02 force gauge/100 mm arm; thermal probe; clutch adjustment tools.
TOOLS Force gauge, angle gauge, caliper, camera

A / PHONE FIT - POWER OFF, CLUTCH DISENGAGED



B / CLUTCH BREAKAWAY TEST - PHONE REMOVED



Restrain the other clutch side with a fixture, never fingers.

1. With power isolated and clutch disengaged, install the stationary USB half first, then the moving half. Support both halves while fitting shoes. Hand-sweep through the intended range; verify no display contact, cable tug, clamp shift or forced hinge path.
2. Record radial carriage position and tangential float at 0, 45, 90, 135 and 180 degrees only where freely reachable. Preserve >=2 mm radial end reserve. Record actual safe endpoint angles rather than forcing mathematical 0/180.
3. Measure phone hinge load and empty-fixture gravity/friction separately. With the phone removed, set the clutch using the restrained fixture and calibrated 100 mm arm: torque = force x 0.100 m. Start near 0.20 N m only if compatible with measured loads; never exceed the 0.50 N m design ceiling.
4. Repeat breakaway measurements in both directions, initially targeting +/-10% repeatability. Lock/mark adjustment. Select the minimum successful current with measured protective margin; current is not a torque calibration. If limits are inadequate, revise the mechanics.

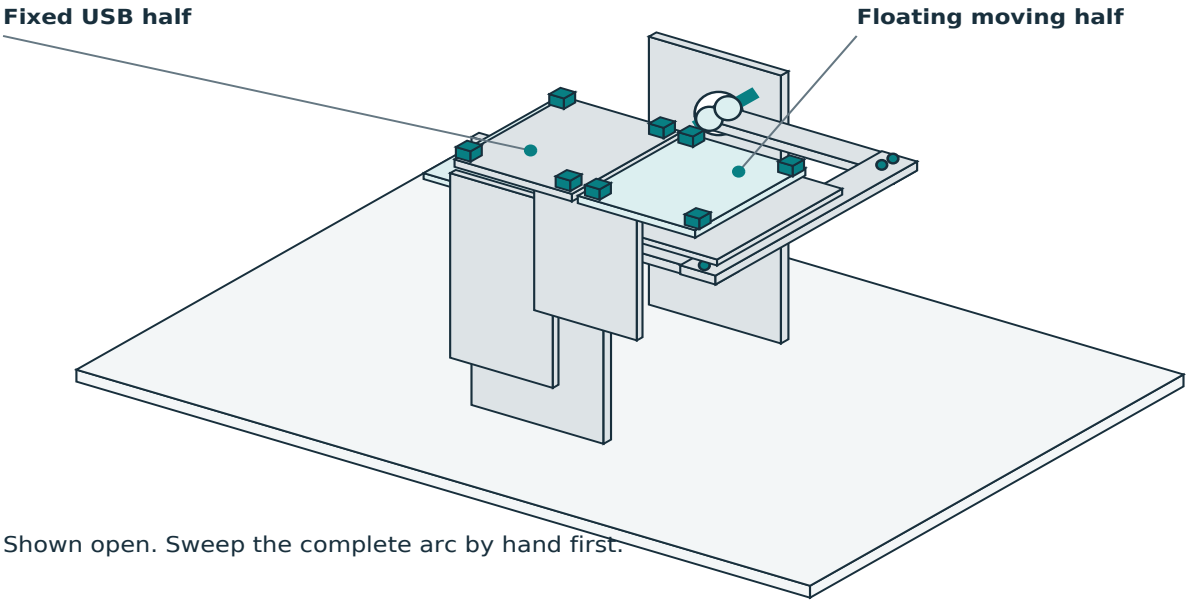
CHECK BEFORE CONTINUING

Phone-specific fit and force record completed. Neither 0.20 N m nor 150 mA is presumed safe for every specimen.

Clutch CW/CCW torque: _____ Fit/load record: _____

Run the first phone trial and release

PARTS Phone-specific supervised-trial record; calibrated bench; API client; inspection log and baseline photographs.
TOOLS Terminal, camera, touch/display test procedure



Shown open. Sweep the complete arc by hand first.

1. Archive the dummy calibration. Under a signed supervised-trial authorization, install a new phone-specific record with measured endpoints and limits. The software requires commissioned=true for motion; that flag is not evidence of routine-release approval.
2. Start one partial-fold cycle, normally 5-175 degrees, 5 degrees/s and 2-second endpoint dwells, using the measured current limit. Keep the guard closed and an operator at the E-stop. Inspect display, touch, frame, hinge, pads and cable after completion.
3. If acceptable, repeat ten cycles, inspect, then 100 and inspect. Record pass/fail through the API at pauses. Reconcile counted cycles independently; mechanical completion does not certify phone function.
4. Sign the final build record for supervised operation, with inspection intervals and remaining restrictions. Routine inspection starts at every 1,000 cycles. Any new shoes, stops, servo, firmware or safety wiring requires recommissioning; unattended service needs a separate release.

CHECK BEFORE CONTINUING

Build records, as-built wiring, vendor drawings, software commit and backups delivered with the bench. Routine release is separate from temporary trial authorization.

Trial record: _____ Routine release/restrictions: _____

Label both ends

Project wire labels below supplement the power-net IDs in the design guide. Confirm actual terminal and connector markings.

Wire IDs	From / path	To	Size
P12-01	12 V +	DC pole A input	18 AWG
P12-02/03	DC pole A output / F1 output	F1 input / relay 13	18 AWG
P12-04/05	Relay 14 / relay 24	Relay 23 / hub +	18 AWG
P12-00	12 V negative	Hub GND	18 AWG
P24-01/02	24 V + / DC pole B output	DC pole B input / F2 input	22 AWG
P24-03/04	F2 output	Relay B1 and 33 (branch)	22 AWG
P24-00	24 V negative	Relay B2 / opto LED return	22 AWG
SA-1/2/3	S21 -> E-stop A -> guard A	S22 (series chain)	22 AWG
SB-1/2/3	S31 -> E-stop B -> guard B	S32 (separate series chain)	22 AWG
SR-1/2	S33 -> reset NO	S34	22 AWG
SL-1	S11	S12 link for selected R1 circuit	22 AWG
ST-1/2	Relay 34 -> 2 kohm -> LED A	LED K -> 24 V negative	24-26 AWG
PI-01/06	Pi pins 1 / 6	3.3 V / ground signal buses	24-26 AWG
PI-11	Permit emitter node via 1 kohm	GPIO17 / physical 11	24-26 AWG
PI-13/15	Open / closed NO node via 1 kohm	GPIO27 / 22, physical 13 / 15	24-26 AWG
PI-07	Both probe DQ; 4.7 kohm pullup	GPIO4 / physical 7	24-26 AWG

Use one verified optocoupler package pin map with step 17. Pi input nodes each have 10 kohm to Pi ground; each GPIO line has 1 kohm in series. Endpoint COM and opto collector use 3.3 V. The two probe grounds use Pi ground, not the isolated 24 V return.

Choose the length from the joint

Inventory kits contain common lengths, not a fully released bolt schedule. Record each final joint before ordering extras.

Joint	Known quantity / size	Selection rule
Base/feet	4 x M6 at four corner holes	Foot stud/insert depth + plate + washers/lock
Tower/feet/adapters	M6 reference; quantity per released drawing	Bearing and angle patterns from delivered parts
Yoke outer joints	4 x M4 total, two per joint	6 mm arm + 6 mm crossbar + washers + locknut; H1
Fixed/moving plates	Four adjustment slots per plate	M4 shoulder or standard screw as joint requires; H2
Moving shoes	4 x M4 shoulder-screw assemblies	8 bushings, spacers and capture washers; H2 stack
Rails/horn/servo/hubs	Use component-specific screws	No assumed thread, depth or tightening torque
Stop/switch brackets	2 stops + 2 switches, four brackets	M4 bracket slots; switch screws from datasheet
Guard/enclosure	24 corner brackets + panel clips/hardware	Matching M5 slot hardware; H3 determines extras

A bolt must clamp before it bottoms

For a through joint, add all gripped material, washers and the actual locking-nut thickness, then allow the locking element's specified thread protrusion. Example: two 6 mm plates already consume 12 mm before washers/nut; an M4x16 from F02 may be too short. Confirm the stack and procure the correct length. For tapped joints, verify engagement and remaining blind-hole clearance from the component drawing.

Joint ID / qty	Selected fastener	Torque + authority	Initials
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Bring-up reference

Run from the repository with Python 3.12 and Pipenv. See README.md for account, service and network configuration.

```
PIPENV_VENV_IN_PROJECT=1 pipenv sync --dev
pipenv run test
# Simulation: generate a private API key first
export FOLDBENCH_MODE=simulation
pipenv run serve
# Stop service / server before the hardware probe
pipenv run python -m foldbench.probe \
  --port /dev/serial/by-id/ACTUAL_U2D2_ID
# Only after physically uncoupling the motor:
pipenv run python -m foldbench.probe \
  --port /dev/serial/by-id/ACTUAL_U2D2_ID \
  --index-tick 1024 --uncoupled \
  --gpio-chip /dev/ACTUAL_HEADER_CHIP
```

Item	Required installation work
Calibration	File outside live API; archive example; record actual angles/ticks, current, speed and temperature limits.
Hardware mode	FOLDBENCH_MODE=hardware; verify serial, GPIO chip and both probe paths. Stop service before editing configuration.
API key	Set FOLDBENCH_API_KEY to a random secret; authorize X-API-Key in /docs. Keep environment file access restricted.
Trial request	POST /v1/runs; new Idempotency-Key. specimen=actual identifier, cycles=1, speed_deg_s=5, dwell_s=2, current_limit_ma=measured value.
Local limits	Initial design ceilings: 15 deg/s, 150 mA; commissioned values must be justified. Absolute design bounds: 350 mA, 0.50 N m clutch.
Thermal record	Initial case 40 C, motor 55 C, ambient 10-30 C. These are starting choices, not validated phone limits.

The trial flag and record authorize motion, not safety by themselves. Use distinct EMPTY-FIXTURE and PHONE-TRIAL records; sign routine release only after the staged tests pass.

Record the actual results

Station serial: _____ Date: _____ Inspector: _____

Check	Acceptance	Measured result / initials
H1-H5 drawings	All affected interfaces signed/revisioned	
Mechanical alignment	Bearing <=0.2 mm; drive target <=0.1 mm	
Float and retention	+/-6 mm radial; >=2 mm reserve; captive shoes	
Cold continuity	Wire list/polarity matches; supplies isolated	
24 V side / Pi side	No intentional connection across optocoupler	
Guard open; no servo	Hub power absent; permit low	
E-stop pressed; no servo	Hub power absent; permit low	
Close/release only	No power restoration without valid reset	
Valid reset	Hub nominal 12 V; permit >2.0 and <=3.3 V	
Held reset / restart	Cannot defeat monitored-start behavior	
Each safety channel	Independently tested by approved procedure	
Endpoint states	Only correct endpoint high; open wire low	
Probe mapping	Two distinct CRC-valid serials; case attachment	
Uncoupled identity	Model 1020, firmware >=38, ID 1, 57600 baud	

Authorization for supervised empty/dummy motion: _____

Unresolved items / restrictions: _____

Reviewer and date: _____

Release the assembled station

All motion tests use a closed guard and a reachable E-stop. Retain the full commissioning record in docs/COMMISSIONING.md.

Test / record	Measured result / evidence location
Motor stop / gravity settling	Time: _____ Distance/angle: _____
Guard access timing	Access prevented until stopped: _____
USB / Pi process / Pi power loss	Fault/no automatic resume: _____
Endpoint/probe failures + dummy jam	Fault/no false count: _____
100 / 1,000 dummy cycles	Count/wear/temperature evidence: _____
Phone fit at reachable angles	Slide reserve / no contact / no slip: _____
Clutch bidirectional repeatability	CW: ____ CCW: ____ spread: ____ N m
Phone trial 1 / 10 / 100 cycles	Display/touch/frame/hinge inspections: _____
Installed calibration	Record ID: _____ Git commit: _____
Routine supervised release	Reviewer/date: _____

First-run and maintenance handoff

Before every run: test E-stop/guard, inspect pads, fasteners, cable loops, probe attachment and disk space. Initially inspect every 1,000 cycles; record pass before resuming. Check clutch setting, rail lubrication per vendor, witness marks and temperature history. Recommission after changes to shoes, clutch, stops, servo, firmware or wiring.

Routine operating restrictions: _____

As-built drawing and backup location: _____

Unattended operation: NOT RELEASED by this booklet. A separate validated machine/thermal risk assessment and duty qualification are required.

Keep these with the bench

Manufacturer sources checked 4 November 2026. Match their revision and ratings to the delivered components.

R1 Pilz PNOZ X3 20547-EN-10, pp.12-13; wiring authority
https://www.pilz.com/download/open/PNOZ_X3_Operating_Manual_20547-EN-10.pdf

R2 ROBOTIS XM430-W350: model, modes, tick scale and watchdog
https://emanual.robotis.com/docs/en/dxl/x/xm430-w350/

R3 ROBOTIS U2D2: TTL connection and external motor power
https://emanual.robotis.com/docs/en/parts/interface/u2d2/

R4 OMRON D4NS: delivered switch/key mounting and contact data
https://www.ia.omron.com/data_pdf/cat/d4ns_d4ns-sk_ds_e_9_9_csm1235.pdf

R5 Analog Devices DS18B20: powered 1-Wire connection
https://www.analog.com/media/en/technical-documentation/data-sheets/ds18b20.pdf

R6 Raspberry Pi: physical GPIO header and 3.3 V interface
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#gpio

R7 Dynatect Slip-Ease: EAS12 configuration; obtain accepted quote
https://www.dynatect.de/en/product/slip-ease-slip-clutches/

R8 Lite-On LTV-817 family datasheet, package pin functions (manufacturer document hosted by Datasheets.com)
https://www.datasheets.com/lite-on/ltv-817s-b/datasheet.pdf

Project source of dimensions and behavior

hardware/BOM.json and BOM.md; docs/DESIGN_AND_ASSEMBLY.md; hardware/foldbench.scad (envelope only); hardware/calibration.example.json; src/foldbench/models.py, probe.py and settings.py; docs/COMMISSIONING.md; deploy/ and README.md.

All assembly drawings in this companion are original vector illustrations. Editable SVGs and the content source are in docs/assembly/. Regenerate using pipenv run assembly-manual. This publication adds instructions and identifies unresolved interfaces; it does not claim a fabricated, tested or certified machine.